Hw1 : Fairness & Bias in AI

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Libraries & Dependencies

```
#run this once, start of each session.
# !pip install ucimlrepo
```

```
# import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from ucimlrepo import fetch_ucirepo
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
import torch.optim as optim
import os
import torch
from torch import nn
from torch.utils.data import DataLoader, TensorDataset
from torchvision import datasets, transforms
import itertools
```

Understanding the Dataset

```
# fetch dataset
adult = fetch_ucirepo(id=2)

# data (as pandas dataframes)
X = adult.data.features
y = adult.data.targets
```

```
adult.variables
```

	name	role	type	demographic	description	units	missing_values
0	age	Feature	Integer	Age	N/A	None	no
1	workclass	Feature	Categorical	Income	Private, Self-emp-not-inc, Self-emp-inc, Feder...	None	yes
2	fnlwgt	Feature	Integer	None	None	None	no
3	education	Feature	Categorical	Education Level	Bachelors, Some-college, 11th, HS-grad, Prof-...	None	no
4	education-num	Feature	Integer	Education Level	None	None	no
5	marital-status	Feature	Categorical	Other	Married-civ-spouse, Divorced, Never-married, S...	None	no
6	occupation	Feature	Categorical	Other	Tech-support, Craft-repair, Other-service, Sal...	None	yes
7	relationship	Feature	Categorical	Other	Wife, Own-child, Husband, Not-in-family, Other...	None	no
8	race	Feature	Categorical	Race	White, Asian-Pac-Islander, Amer-Indian-Eskimo,...	None	no
9	sex	Feature	Binary	Sex	Female, Male.	None	no
10	capital-gain	Feature	Integer	None	None	None	no
11	capital-loss	Feature	Integer	None	None	None	no
12	hours-per-week	Feature	Integer	None	None	None	no
13	native-country	Feature	Categorical	Other	United-States, Cambodia, England, Puerto-Rico,...	None	yes
14	income	Target	Binary	Income	>50K, <=50K.	None	no

```
adult.metadata
```

```
{'uci_id': 2,
 'name': 'Adult',
 'repository_url': 'https://archive.ics.uci.edu/dataset/2/adult',
 'data_url': 'https://archive.ics.uci.edu/static/public/2/data.csv',
 'abstract': 'Predict whether annual income of an individual exceeds $50K/yr based on census data. Also known as "Census Income"
dataset. ',
 'area': 'Social Science',
 'tasks': ['Classification'],
 'characteristics': ['Multivariate'],
 'num_instances': 48842,
 'num_features': 14,
 'feature_types': ['Categorical', 'Integer'],
 'demographics': ['Age', 'Income', 'Education Level', 'Other', 'Race', 'Sex'],
 'target_col': ['income'],
 'index_col': None,
 'has_missing_values': 'yes',
 'missing_values_symbol': 'NaN',
 'year_of_dataset_creation': 1996,
 'last_updated': 'Tue Sep 24 2024',
 'dataset_doi': '10.24432/C5XW20',
 'creators': ['Barry Becker', 'Ronny Kohavi'],
 'intro_paper': None,
 'additional_info': {'summary': "Extraction was done by Barry Becker from the 1994 Census database.  A set of reasonably clean
records was extracted using the following conditions: ((AAGE>16) && (AGI>100) && (AFNLWGT>1)&& (HRSWK>0))\n\nPrediction task is to
determine whether a person's income is over $50,000 a year.\n",
  'purpose': None,
  'funded_by': None,
  'instances_represent': None,
  'recommended_data_splits': None,
  'sensitive_data': None,
  'preprocessing_description': None,
  'variable_info': 'Listing of attributes:\r\n\r\n>50K, <=50K.\r\n\r\nage: continuous.\r\nworkclass: Private, Self-emp-not-inc,
Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.\r\nfnlwgt: continuous.\r\neducation: Bachelors, Some-
college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th,
Preschool.\r\neducation-num: continuous.\r\nmarital-status: Married-civ-spouse, Divorced, Never-married, Separated, Widowed,
Married-spouse-absent, Married-AF-spouse.\r\noccupation: Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-
specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-
serv, Armed-Forces.\r\nrelationship: Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.\r\nrace: White, Asian-
Pac-Islander, Amer-Indian-Eskimo, Other, Black.\r\nsex: Female, Male.\r\ncapital-gain: continuous.\r\ncapital-loss:
continuous.\r\nhours-per-week: continuous.\r\nnative-country: United-States, Cambodia, England, Puerto-Rico, Canada, Germany,
Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica,
Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala,
Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinadad&Tobago, Peru, Hong, Holand-Netherlands.',
  'citation': None}}
```

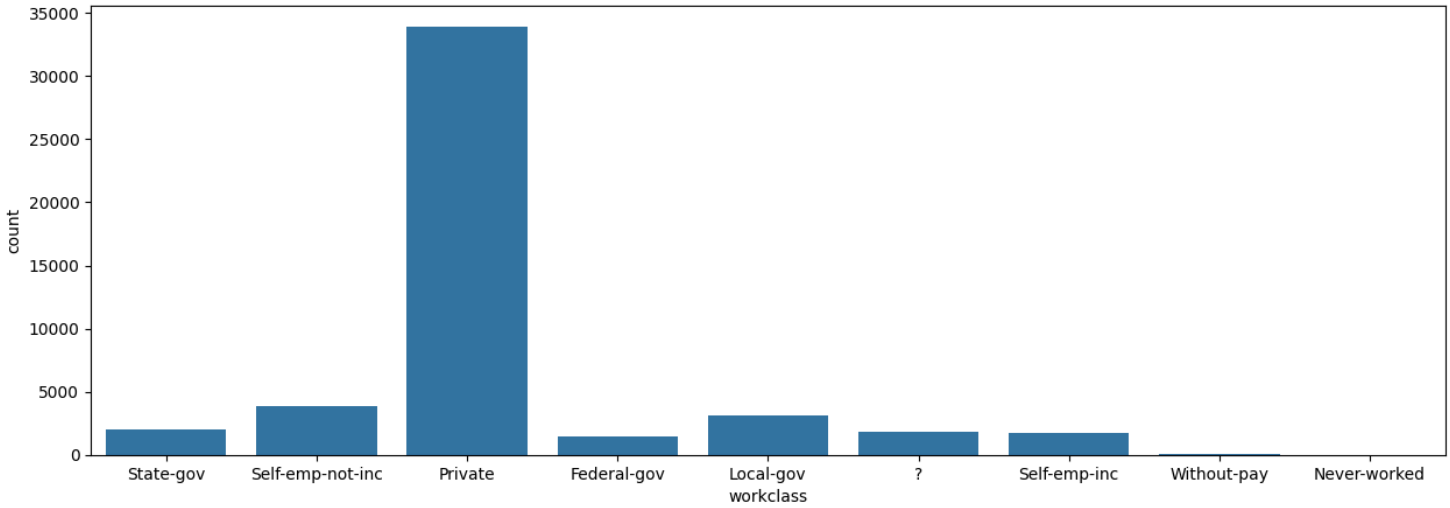

```
df1 = pd.DataFrame(y)
df = pd.DataFrame(X)
df = pd.concat([df, df1], axis=1)
df.head(10)
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0	40	United States
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	13	United States
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40	United States
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0	40	United States
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0	40	Cuba
5	37	Private	284582	Masters	14	Married-civ-spouse	Exec-managerial	Wife	White	Female	0	0	40	United States

Next steps: Generate code with df New interactive sheet

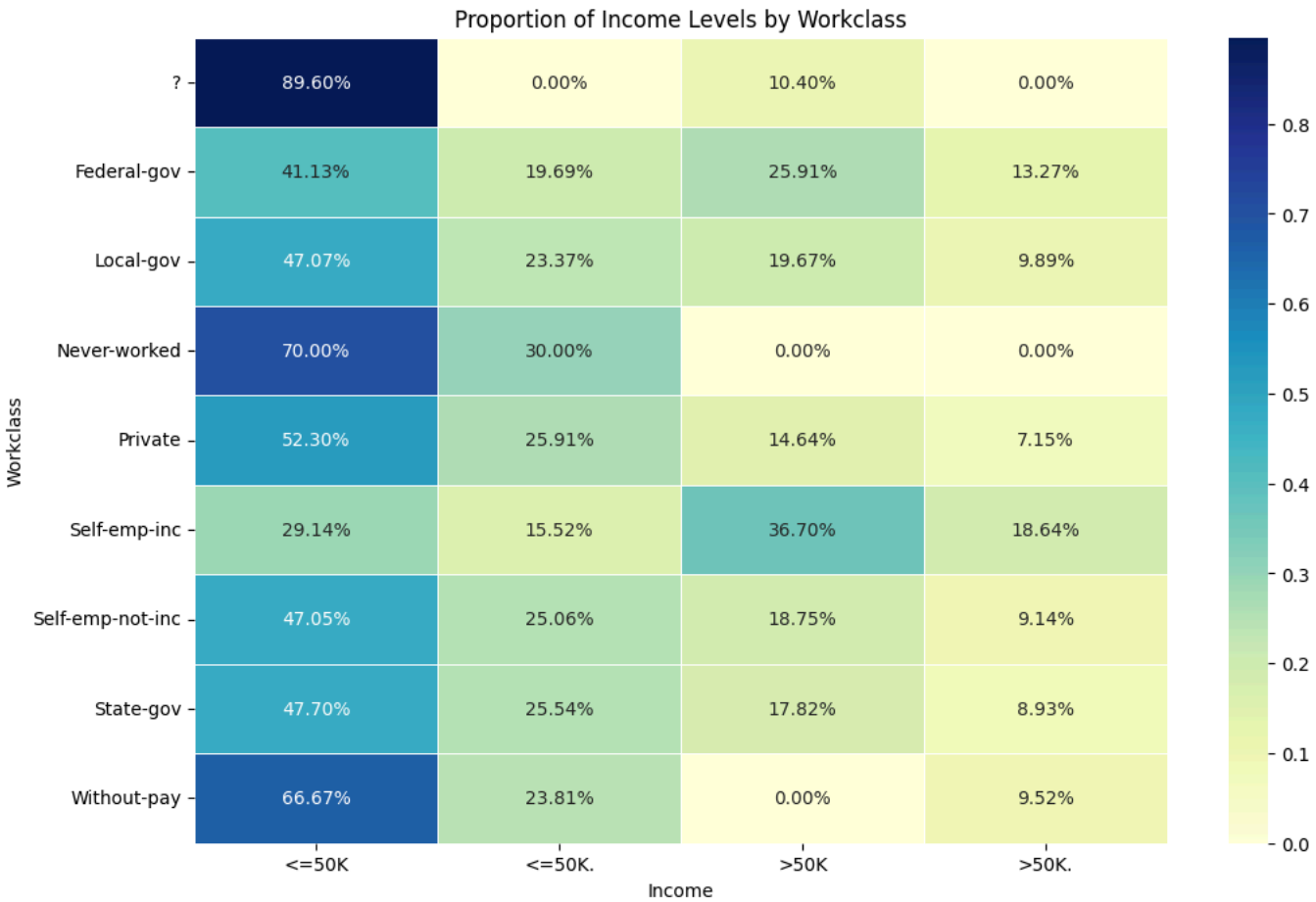
Visualizations

```
plt.figure(figsize=(15,5))
sns.countplot(x='workclass', data=df)
plt.show()
```




```
#contingency table of workclass and income
crosstab_df = pd.crosstab(df['workclass'], df['income'], normalize='index')

plt.figure(figsize=(12, 8))
sns.heatmap(crosstab_df, annot=True, fmt=".2%", cmap="YlGnBu", linewidths=.5)
plt.title('Proportion of Income Levels by Workclass')
plt.xlabel('Income')
plt.ylabel('Workclass')
plt.show()
```




```
df['income'].value_counts()
```

income	count
<=50K	24720
<=50K.	12435
>50K	7841
>50K.	3846

dtype: int64

Fixing Typo and Assigning Binary Labels.

```
df['income'] = df['income'].replace({'<=50K.': 0, '<=50K':0, '>50K.': 1, '>50K':1 })
df['income'].value_counts()
```

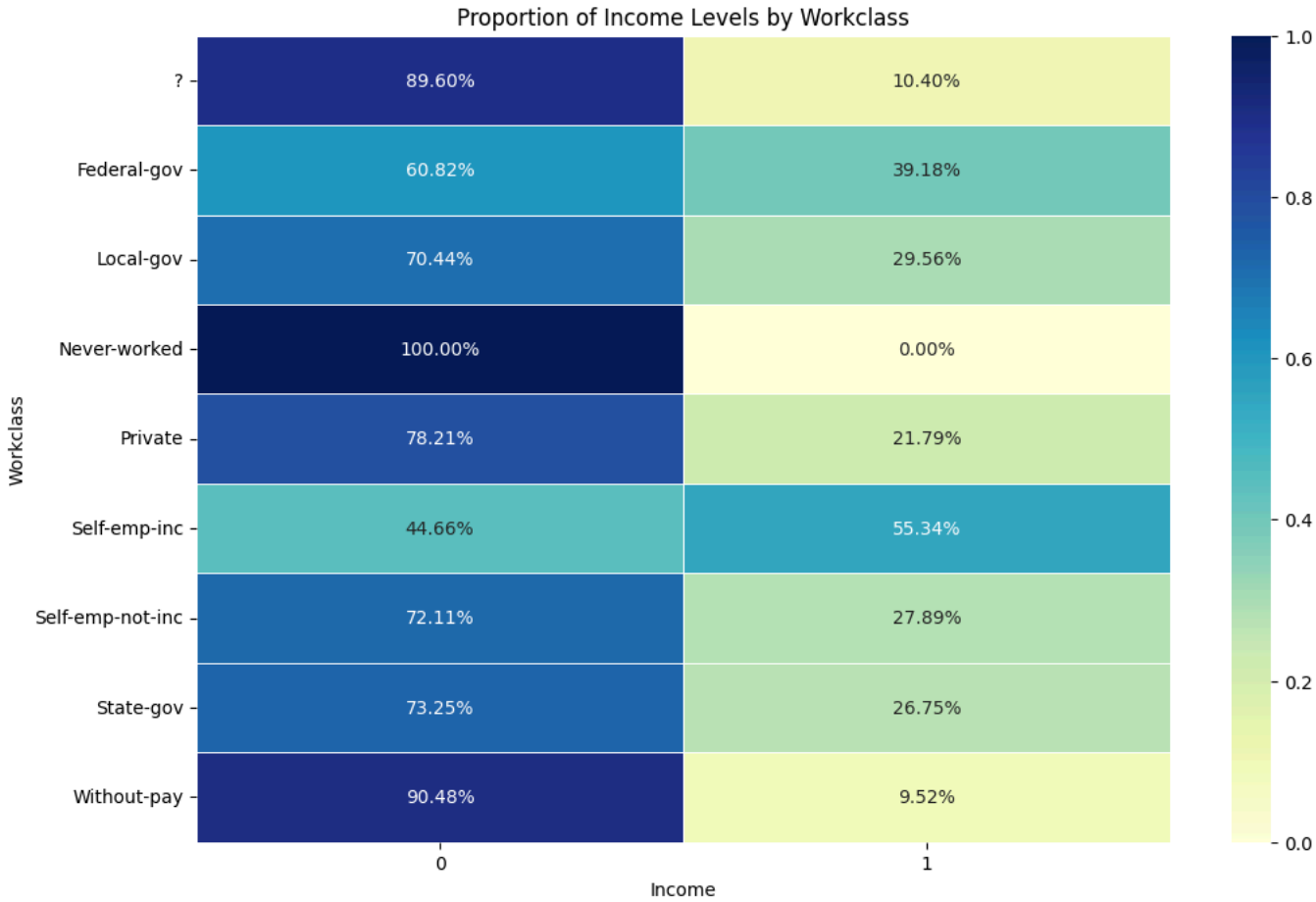
```
/tmp/ipython-input-1855584782.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future
  df['income'] = df['income'].replace({'<=50K.': 0, '<=50K':0, '>50K.': 1, '>50K':1 })
```

income	count
0	37155
1	11687

dtype: int64

```
#contingency table of workclass and income
crosstab_df = pd.crosstab(df['workclass'], df['income'], normalize='index')

plt.figure(figsize=(12, 8))
sns.heatmap(crosstab_df, annot=True, fmt=".2%", cmap="YlGnBu", linewidths=.5)
plt.title('Proportion of Income Levels by Workclass')
plt.xlabel('Income')
plt.ylabel('Workclass')
plt.show()
```




```
Start coding or generate with AI.
```

Preprocessing

```
# dropping null values
df = df.dropna()
X = df.drop('income', axis=1)
y = df['income'].astype(int)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
sex_test = (X_test["sex"].astype(str).str.strip() == "Male").astype(int).values
race_test = X_test["race"].astype(str).str.strip().values

cat_col = X_train.select_dtypes(include=['object', 'category']).columns
print(cat_col)
```

```
Index(['workclass', 'education', 'marital-status', 'occupation',
       'relationship', 'race', 'sex', 'native-country'],
      dtype='object')
```

```
#Chaning Categorical Features to numerical
le = LabelEncoder()

encoders = {}
for col in cat_col:

    train_vals = X_train[col].astype(str).values
    # add UNK to prevent crashes if test contains unseen category
    le.fit(np.concatenate([train_vals, np.array(["UNK"], dtype=object)]))
    encoders[col] = le

    X_train[col] = le.transform(train_vals)

    test_vals = X_test[col].astype(str).values
    known = np.isin(test_vals, le.classes_)
    test_safe = np.where(known, test_vals, "UNK")
```

```
    X_test[col] = le.transform(test_safe)

```

```
# Normalizing the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train.values.astype(np.float32)).astype(np.float32)
X_test_scaled  = scaler.transform(X_test.values.astype(np.float32)).astype(np.float32)

y_train_scaled = y_train.values.astype(np.float32).reshape(-1, 1)
y_test_scaled  = y_test.values.astype(np.float32).reshape(-1, 1)

print("X_train:", X_train_scaled.shape, "y_train:", y_train_scaled.shape,"\n","X_test :", X_test_scaled.shape,  "y_test :", y
print("sex_test:", sex_test.shape, "race_test:", race_test.shape)
print("Categorical col enc:", cat_col)
```

```
X_train: (38096, 14) y_train: (38096, 1)
 X_test : (9525, 14) y_test : (9525, 1)
sex_test: (9525,) race_test: (9525,)
Categorical col enc: Index(['workclass', 'education', 'marital-status', 'occupation',
       'relationship', 'race', 'sex', 'native-country'],
      dtype='object')
```

```
# print(X_train_scaled)
print(X_train.head())
```

```
       age  workclass  fnlwgt  education  education-num  marital-status  \
22858   37          2   48976          9             13               4
47248   23          4  181796         11              9               2
11979   34          4   69251         11              9               2
23082   38          6  184456         11              9               2
1102    39          5  218184          6              5               2

       occupation  relationship  race  sex  capital-gain  capital-loss  \
22858          10             1     5    0          4865             0
47248          12             0     5    1             0             0
11979          14             0     1    1             0             0
23082           8             0     5    1          3464             0
1102            4             0     5    1             0          1651

       hours-per-week  native-country
22858              45              40
47248              50              40
11979              40              40
23082              80              22
1102               40              26
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```

Training

```
def make_loader(X_np, y_np, batch_size=32, shuffle=False):
    X_t = torch.tensor(X_np, dtype=torch.float32)
    y_t = torch.tensor(y_np, dtype=torch.float32)
    ds = TensorDataset(X_t, y_t)
    return DataLoader(ds, batch_size=batch_size, shuffle=shuffle)

train_loader = make_loader(X_train_scaled, y_train_scaled, batch_size=32, shuffle=True)
test_loader  = make_loader(X_test_scaled,  y_test_scaled,  batch_size=1024, shuffle=False)

class MLP(nn.Module):
    # 2 hidden layers (64 each), ReLU; output 1 neuron with Sigmoid
    def __init__(self, input_dim: int):
        super().__init__()
        self.fc1 = nn.Linear(input_dim, 64)
        self.fc2 = nn.Linear(64, 64)
        self.fc3 = nn.Linear(64, 1)
        self.relu = nn.ReLU()
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        x = self.relu(self.fc1(x))
        x = self.relu(self.fc2(x))
        x = self.sigmoid(self.fc3(x))
```

```
        return x

input_size = X_train_scaled.shape[1]
model = MLP(input_size)
print(model)

def probs_to_pred(probs, threshold=0.5):
    return (probs > threshold).astype(int)

@torch.no_grad()
def predict_proba_full(model, X_np):
    model.eval()
    X = torch.tensor(X_np, dtype=torch.float32)
    probs = model(X).detach().cpu().numpy().reshape(-1)
    return probs

def confusion_counts(y_true, y_pred):
    y_true = y_true.astype(int)
    y_pred = y_pred.astype(int)
    tp = np.sum((y_true == 1) & (y_pred == 1))
    fn = np.sum((y_true == 1) & (y_pred == 0))
    fp = np.sum((y_true == 0) & (y_pred == 1))
    tn = np.sum((y_true == 0) & (y_pred == 0))
    return tp, fn, fp, tn

def tpr(y_true, y_pred):
    tp, fn, _, _ = confusion_counts(y_true, y_pred)
    denom = tp + fn
    return float(tp / denom) if denom > 0 else float("nan")

def fpr(y_true, y_pred):
    _, _, fp, tn = confusion_counts(y_true, y_pred)
    denom = fp + tn
    return float(fp / denom) if denom > 0 else float("nan")

def acc(y_true, y_pred):
  accuracy = float(np.mean(y_true.astype(int) == y_pred.astype(int)))
  return accuracy

def equal_opportunity_gap(y_true, y_pred, sex_binary):
    male = (sex_binary == 1)
    female = (sex_binary == 0)
    tpr_m = tpr(y_true[male], y_pred[male])
    tpr_f = tpr(y_true[female], y_pred[female])
    gap = abs(tpr_m - tpr_f)
```

```
MLP(
  (fc1): Linear(in_features=14, out_features=64, bias=True)
  (fc2): Linear(in_features=64, out_features=64, bias=True)
  (fc3): Linear(in_features=64, out_features=1, bias=True)
  (relu): ReLU()
  (sigmoid): Sigmoid()
)
```

```
model = MLP(input_dim=X_train_scaled.shape[1])
optimizer = torch.optim.SGD(model.parameters(), lr=0.01)
criterion = nn.BCELoss()
threshold_default = 0.5

history = {
    "epoch": [],
    "test_acc": [],
    "tpr_male": [],
    "tpr_female": [],
    "gap_eqopp": []
}

y_test_flat = y_test_scaled.reshape(-1).astype(int)

for epoch in range(1, 51):
    model.train()
    total_loss = 0.0

    for xb, yb in train_loader:
        optimizer.zero_grad()
        out = model(xb)
        loss = criterion(out, yb)
        loss.backward()
```

```
        optimizer.step()
        total_loss += loss.item() * xb.size(0)

    probs = predict_proba_full(model, X_test_scaled)
    yhat = probs_to_pred(probs, threshold = threshold_default)

    gap, tpr_m, tpr_f = equal_opportunity_gap(y_test_flat, yhat, sex_test)
    test_a = acc(y_test_flat, yhat)

    history["epoch"].append(epoch)
    history["test_acc"].append(test_a)
    history["tpr_male"].append(tpr_m)
    history["tpr_female"].append(tpr_f)
    history["gap_eqopp"].append(gap)

    if epoch in [1, 20, 50]:
        avg_loss = total_loss / len(train_loader.dataset)
        print(f"epoch {epoch} | train_loss={avg_loss:.4f} | test_acc={test_a:.4f} | "
              f"tpr_m={tpr_m:.4f} | tpr_f={tpr_f:.4f} | gap={gap:.4f}")

print("\n final/last test accuracy : epoch 50 = ", round(history["test_acc"][-1], 5))
```

```
epoch 1 | train_loss=0.4867 | test_acc=0.7965 | tpr_m=0.2606 | tpr_f=0.0535 | gap=0.2071
epoch 20 | train_loss=0.3221 | test_acc=0.8452 | tpr_m=0.5571 | tpr_f=0.4620 | gap=0.0951
epoch 50 | train_loss=0.3141 | test_acc=0.8465 | tpr_m=0.6257 | tpr_f=0.4845 | gap=0.1412

 final/last test accuracy : epoch 50 =  0.84651
```

Start coding or generate with AI.

Part 1:

```
from torch._prims_common import check_all_strides
# Standard dev in eqop gap.
std_gap = float(np.std(history["gap_eqopp"], ddof=0))
print(std_gap)
# T/F
check = (abs(history["gap_eqopp"][49]) < 0.04)
print(check)
```

```
0.05533880733210574
False
```

```
#dummy classifier
majority_class = int(np.mean(y_train_scaled) >= 0.5)  # majority in TRAIN
pred_d = np.full_like(y_test_flat, fill_value=majority_class) #predicting label for every class
gap_d, tpr_m_d, tpr_f_d = equal_opportunity_gap(y_test_flat, pred_d, sex_test)
acc_d = acc(y_test_flat, pred_d)
print(majority_class)
print(f"Dummy accuracy: {acc_d:.4f}")
print(f"Dummy tpr for male: {tpr_m_d:.4f}")
print(f"Dummy tpr for femal: {tpr_f_d:.4f}")
print(f"Dummy gap in equal oppo: {gap_d:.4f}")
```

```
0
Dummy accuracy: 0.7577
Dummy tpr for male: 0.0000
Dummy tpr for femal: 0.0000
Dummy gap in equal oppo: 0.0000
```

Final Answers

Q1) Gap in the equal opportunity in the 1st epoch.

A1) 0.2071

Q2) Gap in the equal opportunity in the 20th epoch.

A2) 0.0951

Q3) Gap in the equal opportunity in the final epoch.

A3) 0.1412

Q4) Standard deviation of the gap in the equal opportunities from epoch 1 to 50.
A4) 0.0553

Q5)True/False: This classifier satisfies equal opportunity with the maximum error of 0.04 (i.e., |TPRmale - TPRfemale|<0.04 , where TPR is True Positive Rate).
A5) False

Q6) If a classifier achieves equal TPR across groups, does it necessarily achieve equal False Positive Rate (FPR)? (Yes/No)? Why ?
A6) No. I don't think so, even if the the TPR is equal among groups it's because Y=1 (might have equal no. of tp) condition allows it, while the FPR depends on Y=0, which is why the tn is independent of the tp and they may or may not be equal for different distributions.

Q7) Train a dummy classifier that always predicts the majority class for all the inputs.What is the gap in equal opportunity for this classifier? Why ?
A7) The gap is zero, the dummy classifier favours the imbalance and always predicts zero, therefore leading to tpr of both male and female to be zero.

```
thresholds = np.arange(0.2, 1.1, 0.2)
final_probs = predict_proba_full(model, X_test_scaled)
metrics = []

for t in thresholds:
    yhat = (final_probs > t).astype(int)
    gap, tpr_m, tpr_f = equal_opportunity_gap(y_test_flat, yhat, sex_test)
    acc_val = acc(y_test_flat, yhat)        #
    metrics.append({"threshold": t, "gap": gap, "acc": acc_val,
                    "tpr_m": tpr_m, "tpr_f": tpr_f})
    print(f"threshold={t:.1f} | acc={acc_val:.4f} | gap={gap:.4f} | "
          f"tpr_m={tpr_m:.4f} | tpr_f={tpr_f:.4f}")


# Q8- first threshold where tpmr_m < tpr_f
for m in metrics:
  if m["tpr_m"] < m["tpr_f"]:
    print("Threshold where TPR of male lesser than female: ", m["threshold"])
    break

```

```
threshold=0.2 | acc=0.7769 | gap=0.1893 | tpr_m=0.9160 | tpr_f=0.7268
threshold=0.4 | acc=0.8383 | gap=0.1673 | tpr_m=0.7419 | tpr_f=0.5746
threshold=0.6 | acc=0.8461 | gap=0.1325 | tpr_m=0.5212 | tpr_f=0.3887
threshold=0.8 | acc=0.8191 | gap=0.0921 | tpr_m=0.2949 | tpr_f=0.2028
threshold=1.0 | acc=0.7577 | gap=0.0000 | tpr_m=0.0000 | tpr_f=0.0000
```

Final Answer

Q8) At what decision threshold does the TPR for the privileged group (male) cross below the TPR for the protected group (female)?
A8) At no decision threshold the True positive rate (tpr) for male goes below the female.

```
best_acc = max(metrics, key=lambda m: m["acc"])
best_gap = min(metrics, key=lambda m: m["gap"])
print(best_acc)
print(best_gap)
check_9 = (best_acc['threshold'] == best_gap['threshold'])
print(check_9)
```

```
{'threshold': np.float64(0.6000000000000001), 'gap': 0.13251696559284026, 'acc': 0.8460892388451443, 'tpr_m': 0.5212493599590374,
{'threshold': np.float64(1.0), 'gap': 0.0, 'acc': 0.7576902887139108, 'tpr_m': 0.0, 'tpr_f': 0.0}
False
```

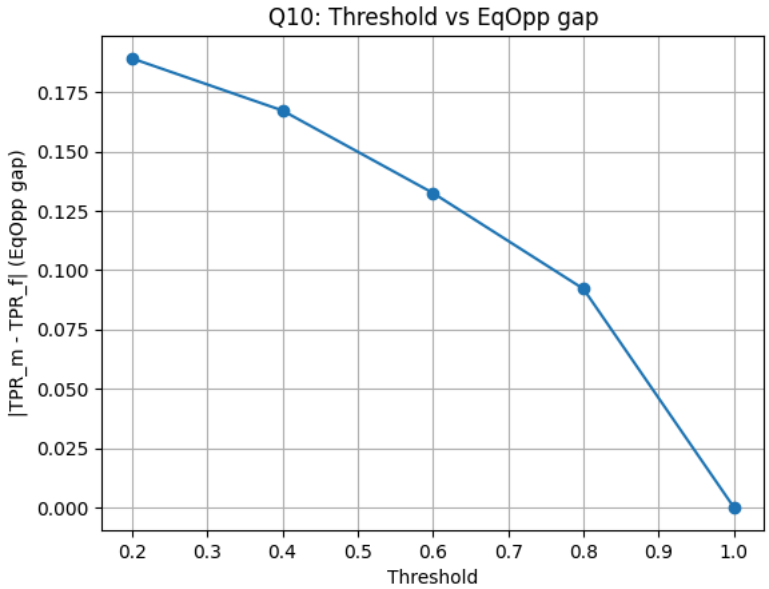
Final Answer

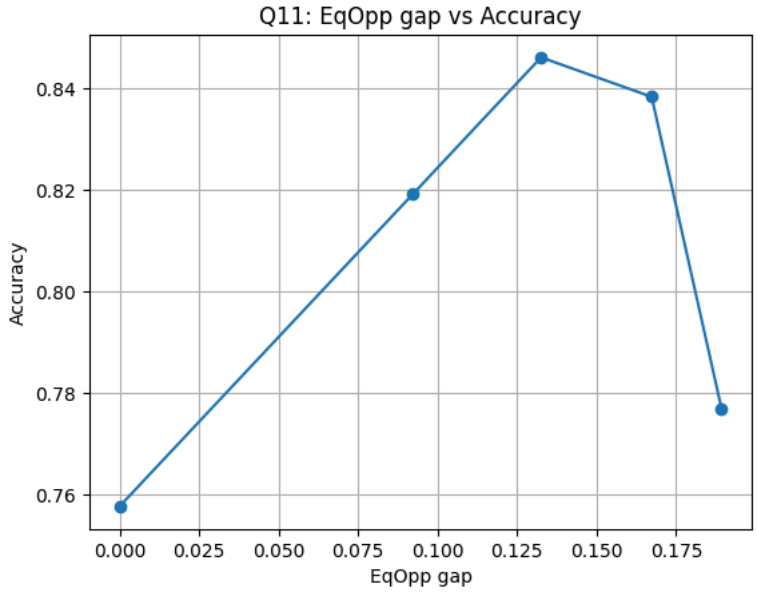
Q9)Does maximizing the accuracy correspond to the threshold with the lowest gap in the Equal Opportunity? (Yes/No) Why?
A)No, both occur at different thresholds. I think that is because higher accuracy is achieved at higher threshold while lowest gap in equal opportunity occurs at extreme threshold levels (highest or lowest). This is becasue the class imabalance favors easier/conservative predictions (0 in our case).

```
print("Final Answer: Q10 and Q11\n\n")
# Q10: Threshold vs gap
plt.figure()
plt.plot([m["threshold"] for m in metrics], [m["gap"] for m in metrics], marker="o")
plt.xlabel("Threshold"); plt.ylabel("|TPR_m - TPR_f| (EqOpp gap)")
plt.title("Q10: Threshold vs EqOpp gap")
plt.grid(True)

# Q11: Gap vs accuracy
plt.figure()
plt.plot([m["gap"] for m in metrics], [m["acc"] for m in metrics], marker="o")
plt.xlabel("EqOpp gap"); plt.ylabel("Accuracy")
plt.title("Q11: EqOpp gap vs Accuracy")
plt.grid(True)
plt.show()
```

```
Final Answer: Q10 and Q11
```





Part2:

```
threshold2 = 0.5
yhat_05 = probs_to_pred(final_probs, threshold=threshold2)
groups = {
    "A_white_male":   (race_test == "White") & (sex_test == 1), #White male
    "B_white_female": (race_test == "White") & (sex_test == 0), #white female
    "C_black_male":   (race_test == "Black") & (sex_test == 1), #black male
    "D_black_female": (race_test == "Black") & (sex_test == 0), #black fem
```

```python
}

group_tprs = {}
for n, g in groups.items():
    group_tprs[n] = tpr(y_test_flat[g], yhat_05[g])
    print(f"tpr: {n}: {group_tprs[n]:.4f}")

# Compute Equal opportunity gap among subgroups
subgroups = []
subgroup_names = list(group_tprs.keys())
for n in range(len(subgroup_names)):
  for o in range(n+1, len(subgroup_names)):
    s1, s2 = subgroup_names[n] , subgroup_names[o]
    equal_opportunity_gap2 = abs(group_tprs[s1]-group_tprs[s2])
    subgroups.append((s1,s2, equal_opportunity_gap2))

for s1,s2, equal_opportunity_gap2 in subgroups:
  print(f"{s1} - {s2} eqop_gap:  {equal_opportunity_gap2:.4f}")

max_pair = max(subgroups, key=lambda x: x[2])
min_pair = min(subgroups, key=lambda x: x[2])

print(f"maximum gap pair: {max_pair[0]} - {max_pair[1]} (eqop_gap={max_pair[2]:.4f})")
print(f"minimum gap pair: {min_pair[0]} - {min_pair[1]} (eqop_gap={min_pair[2]:.4f})")
```

```
tpr: A_white_male: 0.6252
tpr: B_white_female: 0.5196
tpr: C_black_male: 0.6250
tpr: D_black_female: 0.3929
A_white_male - B_white_female eqop_gap:  0.1056
A_white_male - C_black_male eqop_gap:  0.0002
A_white_male - D_black_female eqop_gap:  0.2324
B_white_female - C_black_male eqop_gap:  0.1054
B_white_female - D_black_female eqop_gap:  0.1268
C_black_male - D_black_female eqop_gap:  0.2321
maximum gap pair: A_white_male - D_black_female (eqop_gap=0.2324)
minimum gap pair: A_white_male - C_black_male (eqop_gap=0.0002)
```

Final Answer:

Q1) At the decision threshold of 0.5, what is the TPR for each of the groups?
A1) tpr for each group at threshold = 0.5 is:

- A_white_male: 0.6252
- B_white_female: 0.5196
- C_black_male: 0.6250
- D_black_female: 0.3929

Q2) Based on your answer to the previous question, which of the groups is the most privileged one (A, B, C, D)? why do you think this is the case?
A2) As mentioned in Answer 1, we can see that "Group A: White Male" has the highest true positive rate, Thus they are the most privileged one.

Q3) Calculate the gap in the equal opportunity for all pairs of the subgroups: A-B, A-C, etc. (6 in total). What is the maximum gap in equal opportunity?
A3) Gap in equal opportunity for all subgroups are as follow:

- A-B : 0.1056
- A-C : 0.0002
- A-D : 0.2324
- B-C : 0.1054
- B-D : 0.1268
- C-D : 0.2321

  The, maximum gap pair is 0.2324, among group "A_white_male" and "D_black_female".

Q4) Based on your answer to question 3, for which pair does the maximum gap in the equal opportunity happen? (A-B, A-C, A-D, B-C, B-D, C-D)

A4) As mentioned in answer 3, For Pair A-D there exists the maximun gap in equal opportunity.

Q5) Calculate the gap in the equal opportunity for all pairs of the subgroups: A-B, A-C,etc. (6 in total). What is the minimum gap in the equal opportunity?

A5) The minimum gap in equal opportunity is among the group A-C (white male to black male) and that is of 0.0002.